



Machine Learning Based Battery Second-Life Classification for Sustainable Electric Vehicle Ecosystems

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01 INTRODUCTION

Lithium-ion batteries used in electric vehicles gradually lose capacity and performance during operation. Although many batteries no longer satisfy automotive requirements, they retain sufficient capacity for less demanding stationary energy storage applications. Accurate evaluation of battery health is therefore essential to determine whether a battery should be **reused, repurposed for second-life applications, or recycled**.

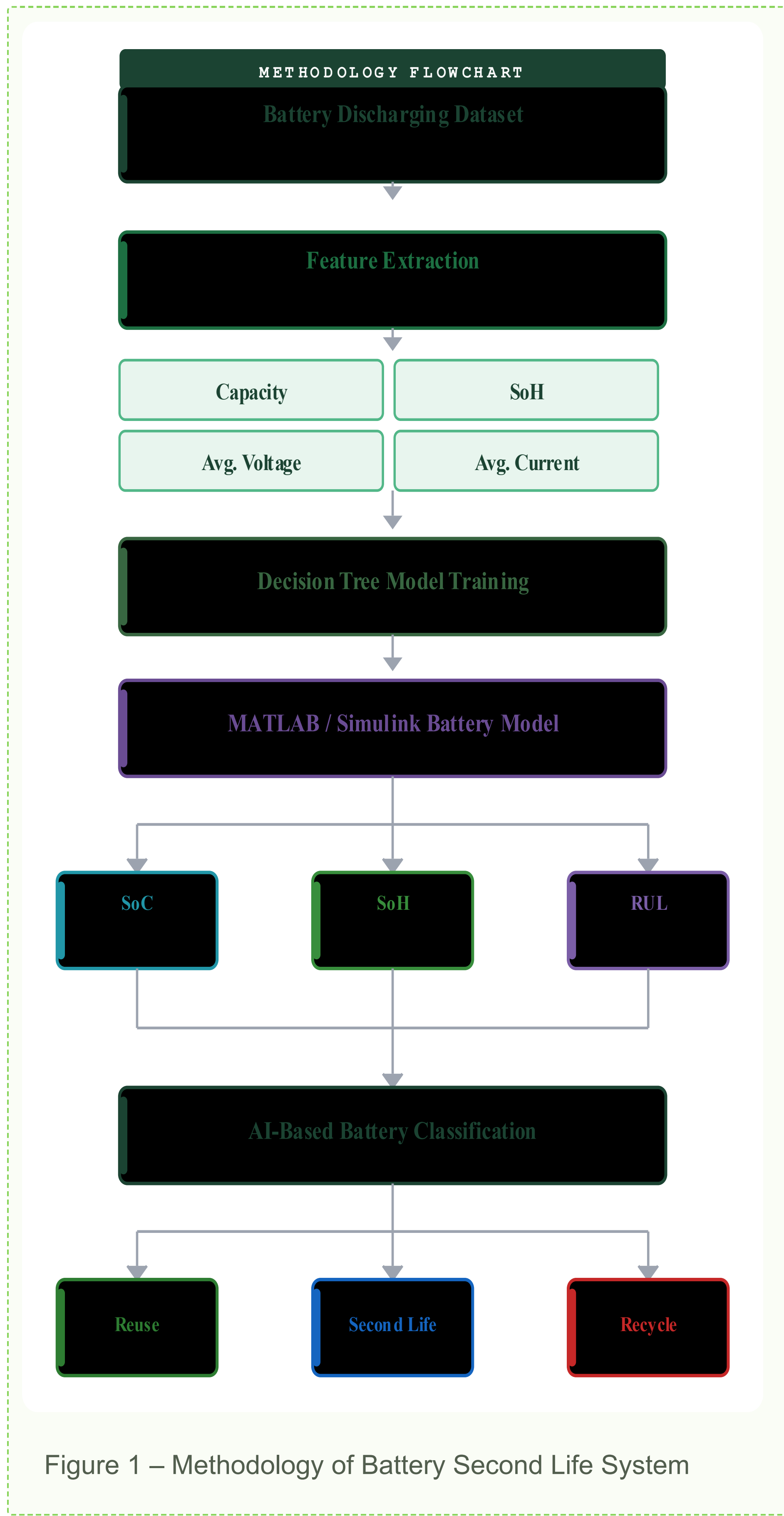
This work presents an **ML-assisted Battery Second-Life Decision System** that integrates **MATLAB/Simulink** with **machine learning** for battery health assessment. State of Charge (SoC), State of Health (SoH), and a preliminary Remaining Useful Life (RUL) are estimated using battery operating data. Features extracted from the **NASA Battery Aging Dataset** are used to train a Decision Tree classifier that categorizes batteries into **Reuse, Second Life, or Recycle**. The proposed framework supports sustainable battery management and contributes to the circular economy by enabling informed end-of-life decisions.

02 OBJECTIVES

- ✓ Develop a MATLAB/Simulink-based lithium-ion battery model to estimate battery State of Charge, State of Health and a preliminary Remaining Useful Life.
- ✓ Build a machine learning framework by extracting features from the NASA Battery Aging Dataset and training a Decision Tree classifier for battery health assessment.
- ✓ Develop an AI-assisted decision system to classify batteries into Reuse, Second Life, or Recycle categories and integrate it with Simulink to demonstrate second life decision framework.

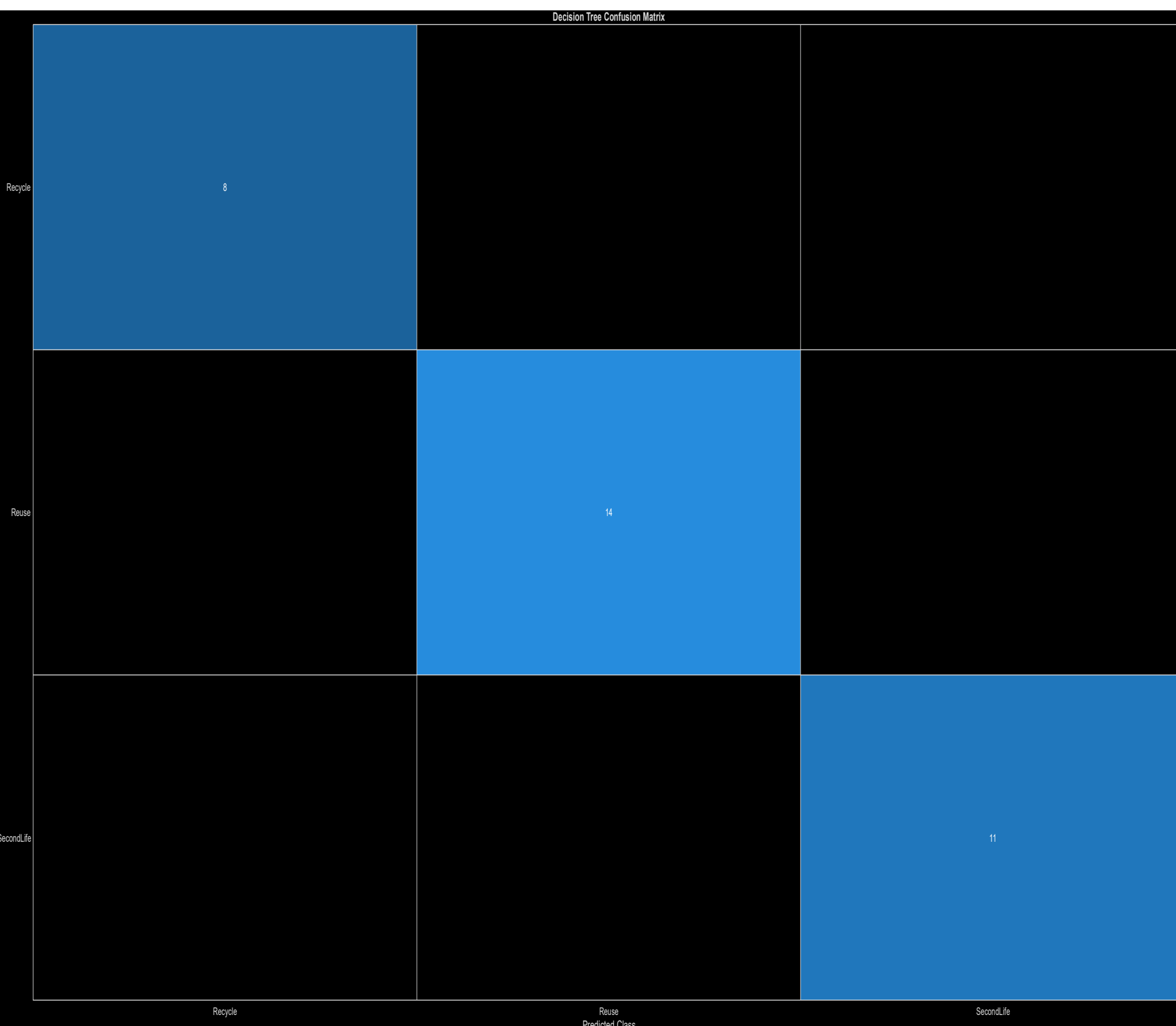
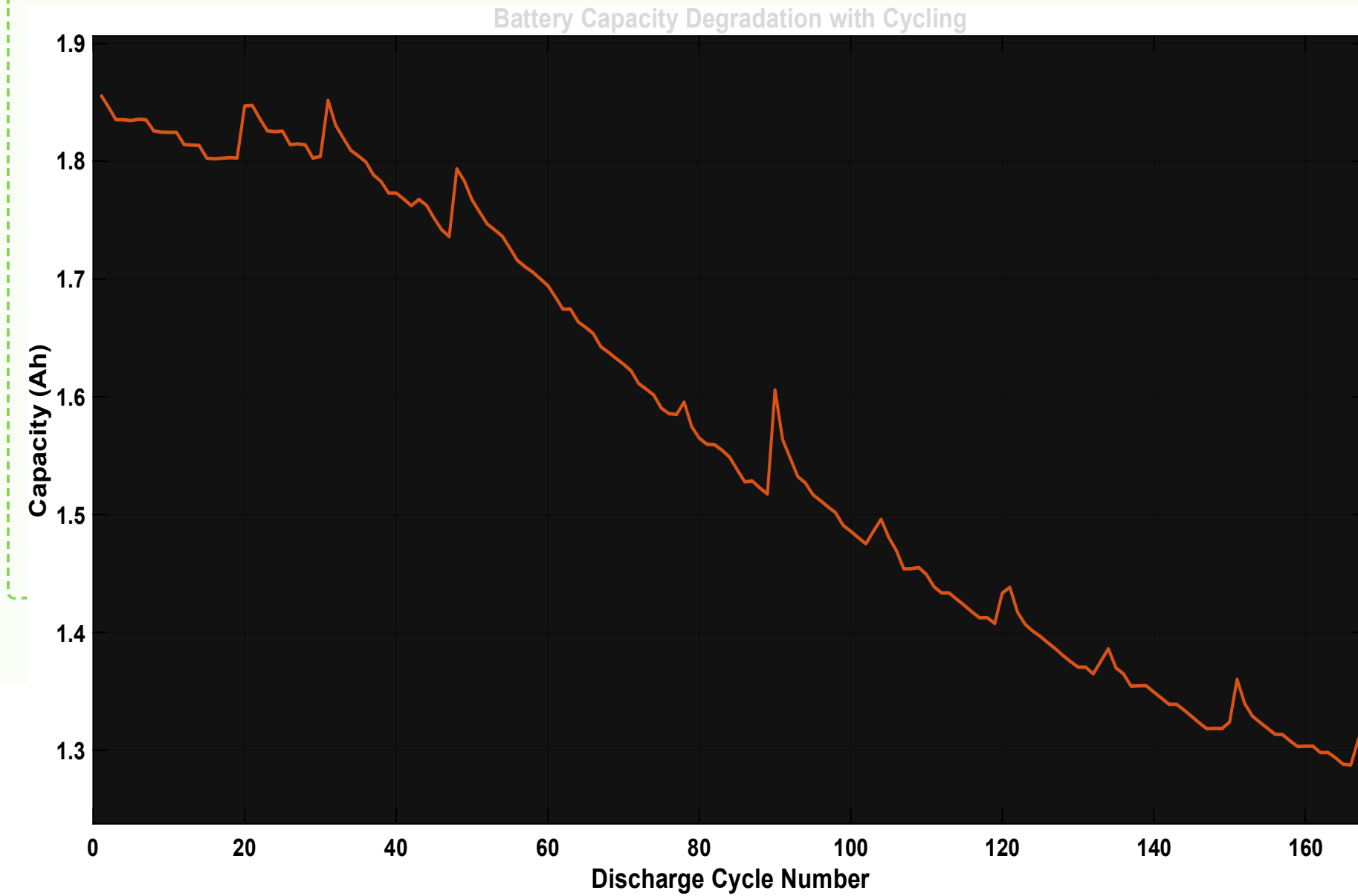
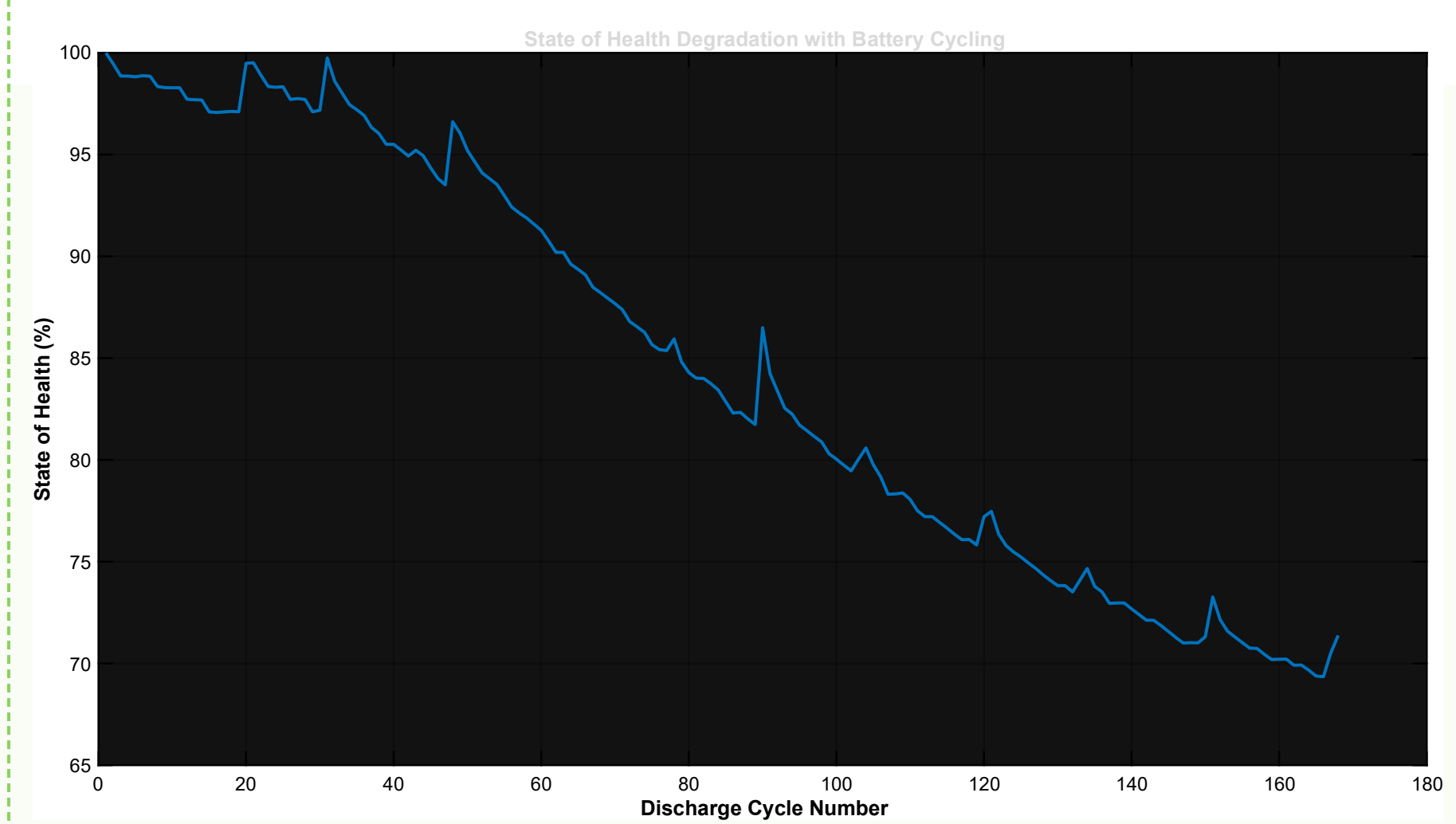
03 METHODOLOGY

A MATLAB/Simulink-based battery model was developed to estimate State of Charge using Coulomb Counting and State of Health using capacity degradation. A preliminary Remaining Useful Life estimation was also incorporated. Battery discharge data from the NASA Battery Aging Dataset were processed in MATLAB to extract features including capacity, SoH, average voltage, and average current. These features were used to train a Decision Tree classifier, which categorizes batteries into Reuse, Second Life, or Recycle. The trained decision logic was integrated into the Simulink model to automate battery second-life decisions.

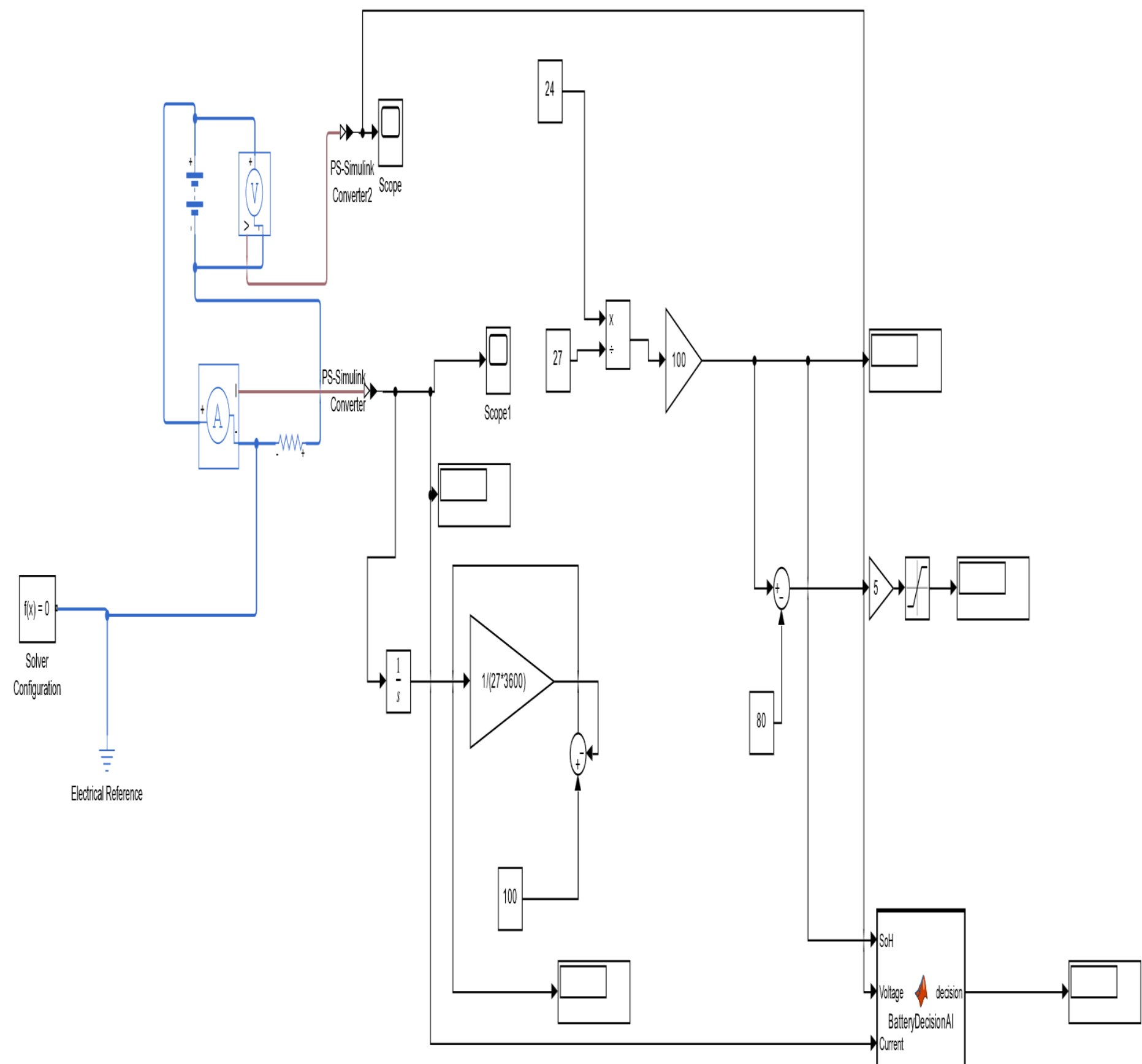


04 RESULTS

The Battery Second Life was successfully implemented in MATLAB/Simulink. Battery Health parameters were estimated using the developed model. Data was extracted from the NASA Battery aging dataset were used to train a Decision Tree classifier. The battery was classified in Reuse, Second Life and Recycle.



05 DISCUSSION



1. The proposed framework successfully integrates battery modelling with machine learning for automated second-life assessment of lithium-ion batteries.
2. The SoH degradation trend obtained from the NASA B0005 dataset demonstrates the reduction in battery capacity with increasing discharge cycles.
3. The Decision Tree classifier effectively categorizes batteries into Reuse, Second Life, and Recycle, supporting sustainable battery lifecycle management. The framework is modular and can be integrated into future Battery Management Systems (BMS) for real-time battery health monitoring.
4. Limitation: The current study is validated using the NASA B0005 dataset and a single Decision Tree model.

06 CONCLUSIONS

- Developed a MATLAB/Simulink framework for battery health
- Classified batteries into Reuse, Second Life, and Recycle using a Decision Tree.
- Promotes circular economy through battery reuse and recycling.

AI-ASSISTED BATTERY HEALTH ASSESSMENT ENABLES INFORMED SECOND-LIFE DECISIONS, SUPPORTING BATTERY CIRCULARITY IN EV ECOSYSTEMS.

07 FUTURE WORK

1. Deploy the proposed framework on an embedded Battery Management System (BMS) for real-time battery monitoring and second-life decision making.
2. Compare the Decision Tree classifier with advanced machine learning algorithms such as Random Forest, Support Vector Machine (SVM), and XGBoost.

08 REFERENCES

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- 2 G. L. Plett, Battery Management Systems, Volume I: Battery Modeling. Artech House, 2015.
- 3 MathWorks. MATLAB and Simulink Documentation. MathWorks Inc.

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